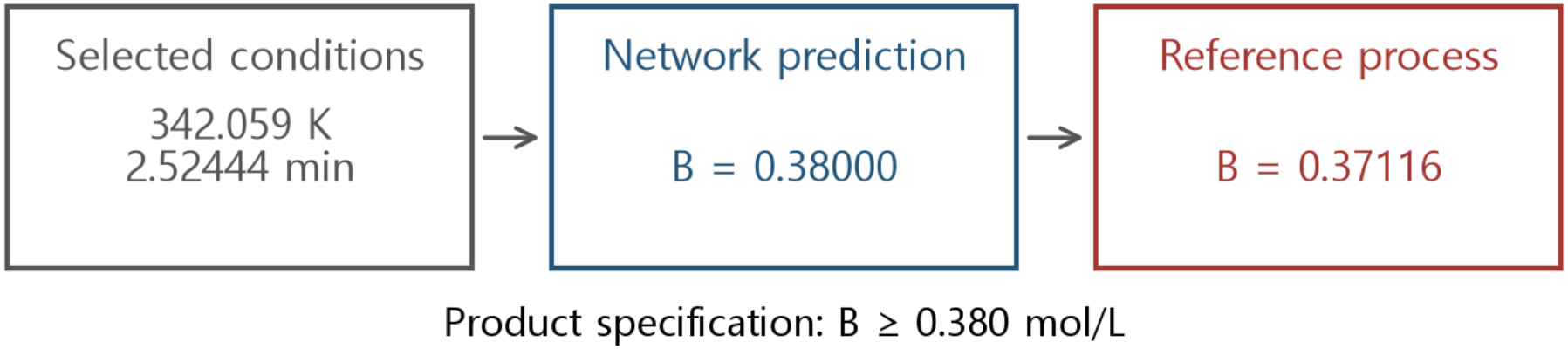
Week 8: Selecting and Revalidating Operating Conditions

Product specifications and revalidation of an optimum

The optimized product concentration meets the specification of 0.380 mol/L. Yet the process equations give 0.371 mol/L at the same conditions. Should this candidate be accepted?



Revalidate the selected conditions, choose an acceptable candidate, and identify the next useful data.

Prediction validation and decision validation

Average error on independent evaluation data summarizes predictive quality across an input region. Optimization selects a few points with low cost, often near constraint boundaries.

Recompute product specifications, physical consistency, and cost at those decisions. Separately check whether the solver correctly represented the fitted model.

This week: these distinctions with the unchanged frozen network from Week 6.

Learning objectives and final report

1. Prediction error, formulation error, and process constraint violation
2. Cost comparison among candidates feasible in the reference process model
3. Revalidation after margins and local data correction
4. Reproducible records of models, data, solvers, and candidate selection

Final report: accepted candidate, rejection reasons, remaining checks, and additional data locations.

Reactor assumptions and variables

Reference process: ideal steady-state isothermal CSTR; consecutive first-order reactions A → B → C. Feed concentrations: A 1.2 mol/L, B and C zero. Constant density and flow.

Temperature T spans 310-350 K and residence time τ spans 2-10 min. At fixed F = 1 L/min, V = Fτ, so τ is also a reactor-sizing decision.

Changing τ in an existing fixed-size vessel requires changing flow or another condition. That requires a revised cost and process formulation.

Temperature-dependent reaction rates

$$k_1\left(T\right) = .2\exp\left[\frac{40000}{8.314}\left(\frac{1}{330} - \frac{1}{T}\right)\right]$$
$$k_2\left(T\right) = .1\exp\left[\frac{50000}{8.314}\left(\frac{1}{330} - \frac{1}{T}\right)\right]$$

Rate constants have units min^{-1}, activation energies are J/mol, and R = 8.314 J/(mol·K). At 330 K, $k_1 = 0.2$ and $k_2 = 0.1$.

Temperature is externally maintained; this model includes no energy balance or cooling-capacity limit.

Unlike the 6000 K and 5000 K sensitivities in Weeks 1–4, this reference uses E/R=40000/8.314 and 50000/8.314 K. Rates agree at 330 K but differ elsewhere. Revalidate the Week 6 weights with the reference used in Weeks 6–8.

Reference process equations and product specifications

$$C_A = \frac{1.2}{1 + k_1\tau}$$

$$C_B = \frac{k_1\tau C_A}{1 + k_2\tau}, C_C = k_2\tau C_B$$

Specifications are $C_B \geq 0.38$ and $C_C \leq 0.25$ mol/L. Concentrations must be nonnegative and sum to 1.2 mol/L.

Revalidation with these equations is conditional on their assumptions. Actual equipment requires additional checks on dynamics, heat management, measurement error, and other operating limits.

Calculate A first, then balance B formation against its outflow and onward reaction. C follows from formation out of B. Use this sequence at the identical T,τ when comparing the network with the reference process.

Cost criterion and comparison baseline

$$J(T,\tau) = 12\tau + .24(T - 310)$$

J is measured in $/h. Assume a volume cost of 12 $/(L·h) and, at fixed flow, a heating coefficient of 0.24 $/(K·h). With F = 1, the volume term becomes 12τ.

This cost compares designs meeting product specifications. It includes neither sales revenue nor actual equipment quotations.

Baseline: T = 330 K, τ = 6 min; cost 76.80 $/h.

At flow 1 L/min, τ=6 min means volume 6 L. Volume cost is 12×6=72 $/h and temperature cost is 0.24×(330−310)=4.8 $/h, totaling 76.8 $/h. This baseline is checked for specifications as well as cost.

The surrogate optimization problem

$$\min_{T,\tau} J\left(T, \tau\right)$$
$$\hat{\mathrm{C}} = f_\theta\left(T, \tau\right)$$
$$\hat{\mathrm{C}}_B \geq .38, \hat{\mathrm{C}}_C \leq .25, \hat{\mathrm{C}} \geq 0$$

Input bounds and scaling match Week 6. The objective is twelve times the Week 6 cost index, so the unadjusted model has the same optimal input.

Use the ReLU MILP from Weeks 5 and 6. The Week 7 cost-epigraph LP does not automatically apply to concentration constraints.

Multiplying every objective value by positive 12 preserves ranking, so the optimal input matches Week 6 while the cost unit changes. Specifications are imposed on network predictions; that ranking alone does not establish feasible costs for the reference process.

Separating errors from termination status

Check	Quantities compared
Formulation error	MILP output and network forward output at the same input
Prediction error	Network output and reference process output
Constraint violation	Reference output and product specifications
Optimization gap	Incumbent and valid bound for the fitted model

The unadjusted solve has MILP/forward disagreement below 10^{-12} mol/L. Its reference product concentration still misses the specification.

Average prediction error and decision-point error

Generate 997 inputs uniformly over the full box, separately from training. RMSE is 0.00535 for A, 0.00439 for B, and 0.00527 for C, in mol/L.

At the optimized candidate, B is overpredicted by 0.00884 mol/L. Since its prediction lies exactly on the specification boundary, this error causes a violation.

Report error at the optimized candidate alongside independent average error so that specification violations at the selected point are visible.

Optimization can reduce cost by reducing specification slack until predicted B touches its lower bound. At that point even a small overprediction causes violation. Evaluate the selected point’s error direction and slack separately from domain-average RMSE.

Reference revalidation of the unadjusted candidate

Quantity	Value
T / τ	342.0593 K / 2.52444 min
Predicted B / reference B	0.380000 / 0.371157 mol/L
Reference C	0.178136 mol/L
B shortfall	0.008843 mol/L
Calculated cost	37.9875 $/h

Reject this candidate because it violates the product specification. Its lower cost does not make it an improvement over a feasible baseline.

Predicted B slack is zero and overprediction is 0.008843, so reference slack is −0.008843. C retains slack 0.25−0.178136=0.071864 mol/L. Failure of even one required specification rejects the candidate.

Input domain and data support

A candidate inside the training box can still have large error if nearby data are sparse or the response varies rapidly. Being inside the box alone does not guarantee accurate interpolation.

Inspect input ranges, scaled coordinates, distances to nearby training points, and errors near boundaries. An expanded optimization domain needs separate model validation.

Recompute valid formulation bounds for the new domain as well.

Physical consistency and product feasibility

$$r_{bal} = \left| \hat{C}_A + \hat{C}_B + \hat{C}_C - 1.2 \right|$$

$$v_B = \max\left(0, .38 - C_B\right)$$

$$v_C = \max\left(0, C_C - .25\right)$$

Record the concentration sum, negative concentrations, and product specifications separately. An exact sum can coexist with low B; positive concentrations can coexist with excessive C.

The Week 4 equality projection enforces its specified linear equality. It does not guarantee product specifications or accurate nonlinear process responses.

Margins from one-sided errors

$$\hat{C}_B \geq .38 + \delta_B$$
$$\hat{C}_C \leq .25 - \delta_C$$

Overprediction of B and underprediction of C are the adverse directions. A separate calibration grid gives δ_B = 0.02130 and δ_C = 0.01305 mol/L.

These are maxima observed at 961 points, not proven bounds over the continuous domain. Reevaluate the new candidate with the process equations.

B has a lower requirement, so guard against the model predicting too much. C has an upper limit, so guard against predicting too little. Opposite inequality directions explain why one margin is added and the other subtracted.

Calculating one-sided margins

Define $e=\text{prediction}-\text{reference}$. Positive e is adverse for B; negative e is adverse for C.

Quantity	B lower limit	C upper limit
Original specification	Reference $B \geq 0.38$	Reference $C \leq 0.25$
Applied margin	$\delta_B = 0.02130$	$\delta_C = 0.01305$
Required prediction	$\hat{B} \geq 0.40130$	$\hat{C} \leq 0.23695$
At the assumed error limit	$0.40130 - 0.02130 = 0.38$	$0.23695 + 0.01305 = 0.25$

The last row applies when the error limit holds at that input. The maximum over 961 calibration points does not prove a bound at every unseen point. Reevaluate the next candidate with the reference process even after adding margins.

Cost and specifications of the margin candidate

Quantity	Result
T / τ	329.1148 K / 4.66529 min
Reference B / C	0.392935 / 0.174547 mol/L
Reference feasibility	Specifications met
Cost	60.5710 $/h
Savings versus baseline	16.2290 $/h

Using a large error observed anywhere in the domain as a margin can be costly in some regions. Compare the margin with error at the actual selected point.

Savings from the baseline are 76.8000−60.5710=16.2290 $/h. First confirm positive reference slacks: 0.012935 for B and 0.075453 mol/L for C; then report the savings.

Choosing additional data locations

Form a 3 × 3 grid around the unadjusted candidate using offsets −0.06, 0, +0.06 on each scaled axis. Clip to the original input box and obtain nine reference outputs.

Local residual correction: these data, separate from the 997 final-evaluation inputs

Choose points to examine constraint boundaries, the current candidate, and error direction. For physical experiments, include accessible conditions and experimental costs.

A local residual model and restricted domain

$$x_1 = \frac{T - 310}{40}, x_2 = \frac{\tau - 2}{8}$$

$$C^{corr}\left(x\right) = f_\theta\left(x\right) + \beta_0 + \beta_1 x_1 + \beta_2 x_2$$

Fit an affine model to reference-minus-prediction residuals at the nine points. Restrict optimization to their bounding box and apply a 0.002 mol/L margin to both product constraints.

This margin is heuristic. Affine correction preserves the MILP structure but provides no uniform bound on residual error.

The correction target is reference minus prediction. At a point overpredicting B by 0.008843, the desired correction is −0.008843. Reversing the sign would raise an already inflated prediction; check corrected output explicitly at one point.

Local-data coordinates and the regression matrix

Candidate (342.0593 K,2.52444 min) has scaled coordinates [(342.0593−310)/40, (2.52444−2)/8]≈[0.80148,0.06556].

Axis	Scaled offset	Physical offset
Temperature	±0.06	±2.4 K
Residence time	±0.06	±0.48 min

```
design = np.column_stack((np.ones(len(x_local)), x_local))
residual = reference_y - network_y
beta = np.linalg.lstsq(design, residual, rcond=None)[0]
corrected_y = network_y + design @ beta
```

For nine inputs, design is (9,3), the three-component residual is (9,3), and beta is (3,3). Multiplying a row $[1, x_1, x_2]$ by beta gives three corrections. Restrict the next optimization to the fitted region and check remaining error at the new candidate.

Failed revalidation after local correction

Quantity	Result
T / τ	339.8152 K / 2.87093 min
Corrected prediction B	0.382000 mol/L
Reference B	0.376820 mol/L
B shortfall	0.003180 mol/L
Cost	41.6069 $/h

Reject this candidate despite nine new data points and a restricted local domain. Its corrected-model error exceeds the 0.002 margin.

The new residual at the candidate becomes useful follow-up data. Do not report the rejected candidate’s cost as achieved savings.

The corrected model predicts 0.382 while the reference gives 0.376820. Subtracting the reserved margin 0.002 from error 0.005180 leaves shortfall 0.003180. Adding data does not by itself establish that remaining error lies within the margin.

Repairing a candidate with the reference process equations

Because this reference model is inexpensive, fix T = 339.8152 K and directly calculate the smaller τ satisfying B = 0.382. Use the original input range, without the local-correction box restriction.

At τ = 3.08559 min, reference C is 0.199530 mol/L and cost is 44.1828 $/h. All original specifications are met.

Specification feasibility of this candidate is checked directly with the reference equations; it lies outside any guarantee supplied by the local correction model.

From the B specification to a quadratic in residence time

At fixed temperature, k_1 and k_2 are numbers. Set B to target b and multiply by the positive concentration denominator.

Step	Relation
Set B=b	$1.2k_1\tau = b(1+k_1\tau)(1+k_2\tau)$
Expand denominator	$1.2k_1\tau = b[1+(k_1+k_2)\tau+k_1k_2\tau^2]$
Collect terms	$bk_1k_2\tau^2+[b(k_1+k_2)-1.2k_1]\tau+b=0$

At 330 K and b=0.4, this is $0.008\tau^2-0.12\tau+0.4=0$. Dividing by 0.008 gives $(\tau-5)(\tau-10)=0$. $B\geq 0.4$ holds between the roots, [5,10].

At fixed temperature, cost increases with τ, so check 5 min first. It gives $C=0.1\times5\times0.4=0.2\leq0.25$. At 10 min, C=0.4 violates the cap; the B condition alone is insufficient.

Computing the smaller root and revalidating the candidate

At T=339.8152 K and b=0.382, calculate $a=bk_1k_2$ and $d=b(k_1+k_2)-1.2k_1$. A negative discriminant means there is no real residence time with B=b at this temperature.

```
k1 = 0.2 * np.exp(40000 / 8.314 * (1/330 - 1/T))
k2 = 0.1 * np.exp(50000 / 8.314 * (1/330 - 1/T))
a = b * k1 * k2
d = b * (k1 + k2) - 1.2 * k1
disc = d*d - 4*a*b
tau = 2*b / (-d + np.sqrt(disc))
```

Here $k_1 \approx 0.3047$, $k_2 \approx 0.1693$/min and $\tau \approx 3.08559$ min. This smaller-root formula uses $a>0, d<0, \text{disc} \geq 0$. Check the original τ bounds and reference B,C for each candidate.

Reference $C \approx 0.199530 < 0.25$ and cost is $12 \times 3.08559 + 0.24 \times (339.8152 - 310) \approx 44.1828$ \$/h. Savings from the original baseline are $76.8 - 44.1828 = 32.6172$ \$/h.

Reference-model optimization for comparison

At each T, solve the quadratic equation for the smaller τ giving B = 0.38. Check τ bounds and the C specification, then use 4,001 temperature grid points and a scalar search near the best grid point.

Comparison candidate: T = 344.9116 K, τ = 2.75110 min; cost 41.3920 $/h.

This supplies a feasible benchmark, not a certified global bound over the continuous domain. Report cost differences as observed benchmark differences.

At fixed temperature, cost increases with τ, so first inspect the smallest τ satisfying the B lower limit. Residence-time bounds and the C limit still apply. The smaller root at an arbitrary temperature is not automatically feasible.

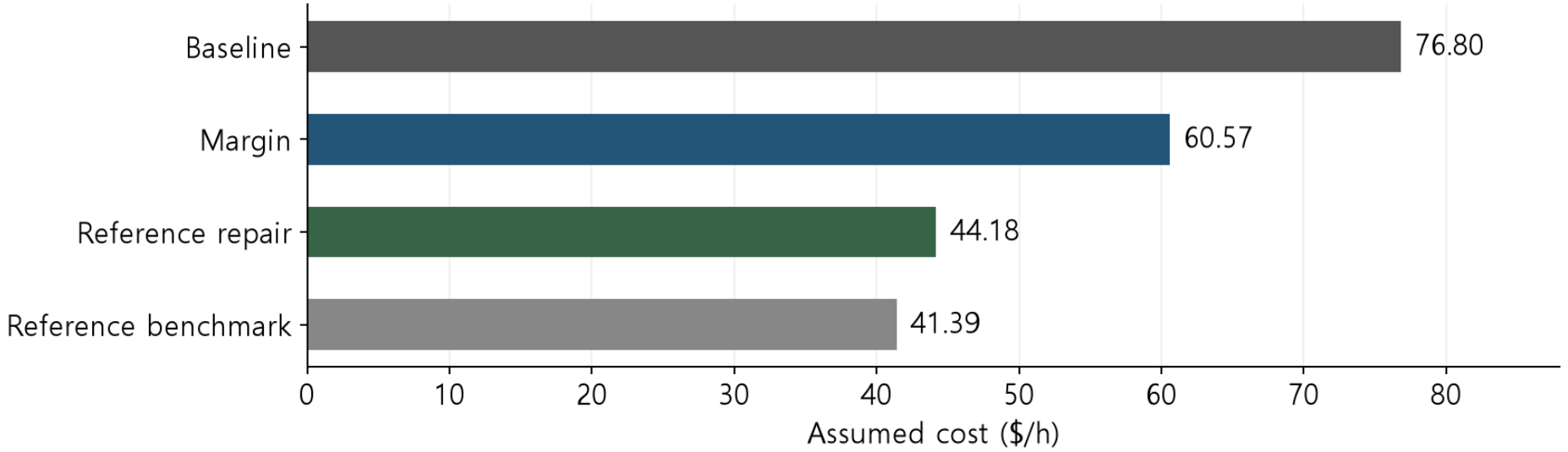
Reference process results by candidate

Candidate	B (mol/L)	Cost ($/h)	Specs
Baseline	0.40909	76.800	Met
Unadjusted MILP	0.37116	37.988	Violated
Margin MILP	0.39293	60.571	Met
Local correction	0.37682	41.607	Violated
Reference repair	0.38200	44.183	Met
Reference benchmark	0.38000	41.392	Met

Every B value uses the same reference process equations. Specification checks include B, C, nonnegative concentrations, and the concentration sum.

Check each row using its complete reference composition. Although the table displays B, also evaluate C, the sum, and nonnegativity. A failed row’s low cost is excluded from rankings of feasible savings.

Cost comparison among feasible candidates



The reference-repaired candidate costs 32.6172 $/h less than the initial baseline, with an observed benchmark difference of 2.7908 $/h.

All values depend on the assumed cost coefficients. Record reference evaluations, model training, and optimization time separately.

Subtracting benchmark 41.3920 from repaired-candidate cost 44.1828 gives 2.7908 $/h. This compares two checked candidates. Without a proof that the benchmark is globally optimal, it is not certified regret relative to the global optimum.

Iteration procedure and acceptance criteria

1. Fix the fitted model, domain, context, specifications, and cost.
2. Obtain a candidate and check forward agreement and solver termination.
3. Check original constraints with the reference equations. On failure, retain a feasible incumbent.
4. Add data, revise the model or region, and revalidate the next candidate.

Convergence of trust-region methods requires conditions on model accuracy and acceptance rules. Those conditions have not been established for this local correction procedure, so convergence is not guaranteed.

Alexandrov, Dennis, Lewis and Torczon, A trust-region framework for managing the use of approximation models in optimization (1998), author manuscript

Reproducibility items in the final report

Category	Record
Model/data	Frozen-weight hash, scaling, training/calibration/evaluation/update data
Problem	Input bounds, fixed context, units, objective, and constraints
Computation	Solver/version/status/gap/time and forward error
Decision	Reference output, violations, cost, and acceptance by candidate
Remaining checks	Reference assumptions, dynamics, heat management, and experiments

Running the lab and report assignment

```
python -m pip install -r requirements.txt
python week08_lab.py --output-dir my-results
```

The bundle includes the frozen Week 6 network and required calculation functions. It saves a results JSON, candidate-comparison CSV, independent evaluation data, and nine update points.

Change a margin or local-grid width once, then revalidate the new candidate against the original specifications. Record data count and cost and explain acceptance or rejection.

Companion and primary sources

Week 8 companion

Week 6 lecture · Week 6 model and executable lab

Alexandrov, Dennis, Lewis and Torczon, A trust-region framework for managing the use of approximation models in optimization (1998), author manuscript

OMLT: Ceccon et al., JMLR 23(349), 2022 · HiGHS